Question

Treatment of NSF_3 with BCl_3 yields an ionic compound $\mathbf{A}$ containing N, B, S, and Cl. In $\mathbf{A}$, the ratio of N to B is 1:1. The theoretical mass fraction of N in the cation is 9.40% (relative atomic masses are retained to two decimal places), and the anion is a tetrahedral ion. The decomposition of $\mathbf{A}$ produces two liquids with the same elemental composition but different colors at room temperature. Regarding the decomposition reaction, which of the following statements is correct?

A. There are a total of four products; the sum of the stoichiometric coefficients after balancing the reaction equation is 8; elemental substances exist among the products.

B. There are three products in total; the sum of stoichiometric coefficients after balancing the reaction equation is 8; elemental substances are present among the products.

C. There are four types of products in total; the sum of stoichiometric coefficients after balancing the reaction equation is 4; no elemental substance exists among the products.

D. There are three types of products in total; the sum of stoichiometric coefficients after balancing the reaction equation is 4; elemental substances exist among the products.

Answer

Correct Answer: A

Detailed Explanation

Since there is no F element, the most probable anion is BCl_4^-.

CHECKPOINT 1 PTS

The anion is BCl_4^-.

Assuming the presence of one N in the cation, after removing N, the remaining mass is 135.03, which may correspond to S and Cl elements. Through simple enumeration, it can be found that this precisely matches 2S and 2Cl.

CHECKPOINT 1 PTS

Enumeration calculations show that $\mathbf{A}$ with 2S + 2Cl excluding the N atom best fits the mass fraction.

Thus, $\mathbf{A}$ is $[N(SCl)_2][BCl_4]$.

The two liquids with identical elemental compositions correspond to the classic SCl_2 and S_2Cl_2. In a decomposition reaction, it is unlikely for N to form compounds with B and Cl as products; instead, N_2 should be produced, while B can only combine with Cl to yield BCl_3.

CHECKPOINT 1 PTS

The products are SCl_2, S_2Cl_2, N_2, and BCl_3

Therefore, the equation is: $2[N(SCl)_2][BCl_4] = N_2 + S_2Cl_2 + 2SCl_2 + 2BCl_3$, and the answer is A.

CHECKPOINT 1 PTS

$$2[N(SCl)_2][BCl_4] = N_2 + S_2Cl_2 + 2SCl_2 + 2BCl_3$$